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Spatial variation of soil nutrients on sandy-loam soil

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ABSTRACT

The spatial variability of plant available phosphorus, plant available potassium, soil pH and soil organic matter content in central Croatia was investigated using geostatistical tools and geographical information system to create nutrient maps and provide useful information for the application of inputs that will also be used for the design of an adequate soil sampling scheme. In a regular grid (50 m × 50 m), 330 samples were collected on sandy loam Stagnic Luvisol. Soil available phosphorus and plant available potassium showed relatively high spatial heterogeneity, ranging from 105 mg kg^{-1} to 310 mg kg^{-1}, and from 115 mg kg^{-1} to 462 mg kg^{-1}, respectively. Content of soil organic matter and pH had lower variability ranging from 1.26% to 2.66% and from 3.75 to 7.13, respectively. Investigated soil properties did not follow normal distribution. Logarithm and Box–Cox transformation were applied to achieve normality. Directional exponential model for soil available phosphorus, potassium and pH and spherical model for soil organic matter was used to describe spatial autocorrelation. Fourteen different interpolation models for mapping soil properties were tested to compare the prediction accuracy. All models gave similar root mean square error values. Available phosphorus, potassium and pH evaluated by radial basis function models (CRS, IMTQ and CRS, respectively) provide a more realistic picture of the structures of analyzed spatial variables in contrast to kriging and inverse distance weighting models. For soil organic matter datasets the most favorable model was LP1. According to the best model soil nutrient maps were created to provide guidance for site-specific fertilization and liming. Soil fertility maps showed sufficient concentrations of soil available phosphorus and available potassium. Acidity map showed that the largest part of the investigated area is very acid and acid. For future management it is necessary to provide more liming materials while fertilization rate should be lower.

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1. Introduction

Nowadays, the vast majority of farmers in Croatia use fertilizers without analyzing the soil fertility. Just a minority of farmers use fertilizers according to soil analysis but without considering in-field spatial variability. In recent decades better investigation methods resulted in higher awareness of crop and soil science and soil properties variability within fields. Most of the yield variability studies have focused on soil nutrient availability and less on soil physical property variability (Castrignano et al., 2002). In contrast to texture and soil organic matter content, nutrients (e.g., potassium, phosphorus) have higher temporal change in soils depending on soil or crop property (Heege, 2013). Therefore, it is necessary to monitor the condition of the soil in a shorter period of time which raises the operating costs. Present technology allows us to improve the application of fertilizers and other inputs by varying rates and blends as needed within fields (Robert, 2002). Geostatistical concept with properly applied sampling trough interpolation technique allows us to assess the value of a soil property at unsampled locations within the area covered by observation points. Respecting the principles of regionalized variable theory (Clark, 1979; Journel and Huijbregts, 1978; Matheron, 1963) it is possible to find the necessary dependence between two sampling points in space. But mapping the spatial distribution of nutrients needs better spatial interpolation methods and their comparisons. Papers that compare different interpolations methods have already been published (Pereira et al., 2013a,b; Robinson and Metternicht, 2006; Xie et al., 2011; Yang et al., 2004; Yasrebi et al., 2009) but the results are not clear. Studies reported different distances between dependent sampling points. Fu et al. (2010) found ranges of 264 m, 300 m and 290 m for phosphorus, potassium and pH on grassland. Robinson and Metternicht (2006) in topsoil of dryland paddock reported ranges of 2110 m

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and 273 m for pH and organic matter, while Kravchenko and Bullock (1999) recorded spatially dependent sampling points from 82 m to 608 m for phosphorus and 85 m to 532 m for potassium in 30 different agricultural fields. Montanari et al. (2012) reported different spatially dependent ranges for phosphorus depending of soil type. Two ranges were 1419 m on Alfisols and 689 m on Alfisols and Oxisols. On colluvial and alluvial soils in regular grid 500 m × 500 m Sağlam et al. (2011) recorded spatial dependent ranges for soil organic matter between 809 and 1680 m depending on soil type. Some researchers prefer ordinary kriging (Kravchenko and Bullock, 1999; Laslett et al., 1987), others (Gotway et al., 1996; Weber and Englund, 1992) inverse distance weighting (IDW) and splines (Robinson and Metternicht, 2006) depending on the investigated parameters, distance between sampling points, dealing with data transformation and the number of neighborhood data points used. Furthermore, interpolation is always associated with some degree of uncertainty, because estimation of an unsampled point produces some error, independent of the method used (Ahmed et al., 2011). The purpose of this study was to investigate statistical methods to describe the observed spatial patterns of soil available potassium (AK), available phosphorus (AP), soil acidity (pH) and soil organic matter (SOM). By comparing the interpolation accuracy for each method and analyzing the errors, the objective of this study was to try to determine the optimum method for analyzing spatial variations of investigated properties.

2. Materials and methods

2.1. Study site, soil sampling and analysis

The study area is located near Zagreb (Croatia) at 45° 31′ N and 16° 59′ E and 112 m.a.s.l. in a rural/forest interface area. Soil sampling was performed in a period from 23 to 26 July 2012, after harvest of winter wheat at the field of 84 ha. Soil samples (0–30 cm) were taken (330) with the newly designed soil sampling probe (for more information see Mesic et al., 2013) in a 50 m × 50 m grid. The location of the sampled field is shown in Fig. 1. Precise location of sampling at grid intersections was set up using Trimble GeoExplorer GeoXH 6000 with ±10 cm accuracy. Sampling, transport and preservation of samples were conducted in compliance with the protocol ISO 10381 from 1 to 8 (1993–2006) (HRN ISO 10381-6, 1993; HRN ISO 10381-3, 2001; HRN ISO 10381-1, 2002; HRN ISO 10381-2, 2002; HRN ISO 10381-4, 2003; HRN ISO 10381-5, 2005; HRN ISO 10381-7, 2005; HRN ISO 10381-8, 2006) and Mesić et al. (2008). Preparation of samples for

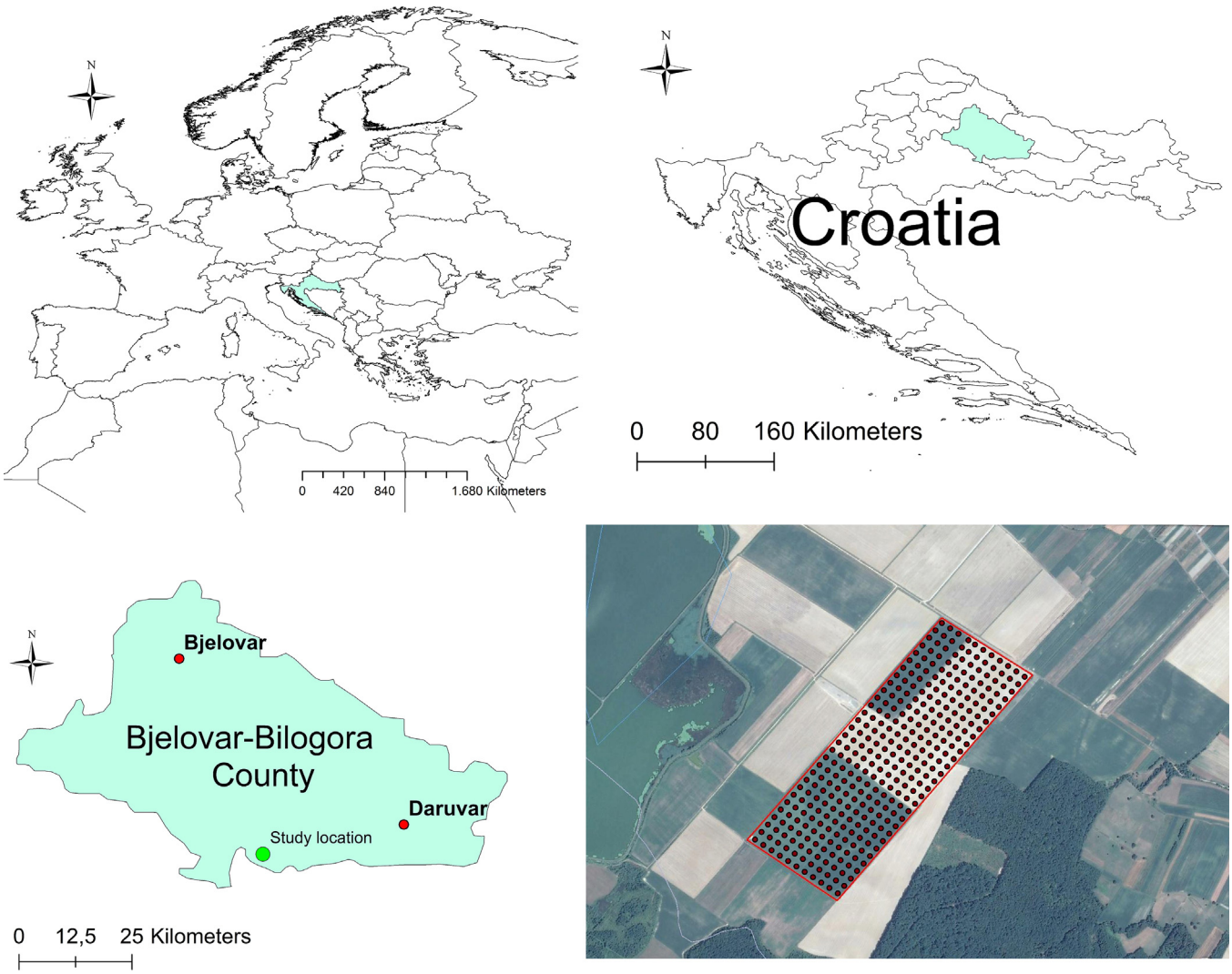


Fig. 1. Study area in Moslavina region, Central Croatia.

chemical analysis (drying, milling, sieving, homogenizing) was conducted in compliance with protocol HRN ISO 11464 (2004). Soil pH was measured using the electrometric method in 1:5 (w/v) ratio with the Beckman pH-meter Φ72, in Kl suspension (HRN ISO 10390, 2005). AP and AK were extracted by ammonium lactate (AL) solution (Egner et al., 1960) and detected by spectrophotometric and flame photometric, respectively. In all soil samples mass ratio of SOM was analyzed using dry combustion method (HRN ISO 10694, 2004) with the vario MACRO CHNS analyzer in the Analytical laboratory of the Department of General Agronomy, Faculty of Agriculture University of Zagreb.

The soil is classified as Pseudogley according to Croatian classification (Škorić et al., 1973) or Stagnic Luvisols according to WRB classification (FAO, 2006). By its texture the soil on the experimental field belongs to sandy loam (Table 1). Climate is semi humid to humid with annual precipitation (in referent period 1960–1991) of 863 mm and average annual temperature of 10.7 °C (Meteorological and hydrological Service of Croatia – meteorological station in Daruvar, 15 km away from investigated area).

Usual crop rotation in that area is winter wheat, maize, soybean, barley and oilseed rape. Soil sampling was performed after winter wheat harvest.

2.2. Statistical analyses and data transformation

Some basic statistics were analyzed, such as minimum (min), 5th percentile (5%), 1st quantile (25%), median, 3rd quantile (75%), 95th percentile (95%), maximum, mean values and the coefficient of variation (CV). The descriptive parameters were calculated using Microsoft Excel for Windows. The Kolmogorov–Smirnov (K–S) test and *Q*–*Q* plots were created to analyze the data distribution and potential outliers using SAS software (Version 9.3). The K–S method, the skewness and kurtosis evaluate data normality and asymmetry that have important implications on interpolation methods performance.

Data were subjected to quality control before modeling. In order to achieve a correct interpretation of the spatial interpolation, it is desirable to have data as close as possible to the normal distribution (Clark and Harper, 2000; Goovaerts et al., 2005; McGrath et al., 2004). High skewness and the presence of outliers can have negative consequences on the accuracy and semivariogram interpretation. To avoid these problems raw data were tested and transformed when necessary. The most common methods were logarithmic (log) and Box–Cox (BC) transformations used to normalize data sets (Fu et al., 2010; Thayer et al., 2003). Log transformations are commonly used to normalize positive skewed data (McGrath et al., 2004) but this transformation was not enough on some occasions. In order to improve data when log transformation fails, BC transformation is more powerful and widely used method (Box and Cox, 1964; Gallardo and Paramá, 2007).

2.3. Geostatistical analysis

In this study geostatistics was used to describe spatial variation of each investigated soil property. Spatial patterns of nutrients were observed with the experimental semivariogram modeling using the data developed to identify the spatial continuity of AP, AK, pH and SOM among sampling points. Semivariogram assists dissimilarity between paired data values to the distance between each sample pair (Goovaerts, 1997; Webster and Oliver, 2007) and gives the spatial structure of the variable and the correlation with the closest points. There are some frequently used models to describe experimental semivariograms: spherical, exponential, Gaussian and linear (Isaaks and Srivastava, 2011).

Semivariogram can be expressed as:

$$\gamma(h) = \frac{1}{2N(h)}\sum_{i=1}^{N(h)}[Z(xi) - Z(xi+h)]^2$$

where $\gamma(h)$ is the semivariance at a given distance h; $Z(xi)$ is the value of the variable Z at the xi location and $N(h)$ is the number of pairs of sample points separated by the lag distance h.

For spatial data, semivariance increases with the distances that are correlated and if the variogram stabilizes it reaches the sill ($C + C_0$), showing that at distance x the samples are considered spatially independent. The variance observed at a shorter scale than the field sampling is identified at 0 lag distance, known as the nugget effect – C_0. This nugget effect makes the short-scale variability and/or sampling errors evident (Burgos et al., 2006; Cetin and Kirda, 2003; Fu et al., 2010). In this study the modeled semi variograms are directional. According previous studies (Webster and Oliver, 2007; Robinson and Metternicht, 2006) number of required sampling points is higher than minimum (150) to reliably identify the presence of anisotropy which considers that the variability is not equal in all directions. In order to describe the directional semi variogram, direction were tested from 0 to 360 by 5 to achieve highest range with applied angular tolerance of 45°. Variable spatial dependency was calculated by the nug/sill ratio. If the ratio is lower than 25%, the variable has strong spatial dependence; if it is between 25% and 75%, the variable has moderate spatial dependence; and if it is higher than 75% the variable shows only weak spatial dependence (Chien et al., 1997). Normally, strong spatial dependence is attributed to intrinsic factors and weak spatial dependence is attributed to extrinsic factors (Cambardella et al., 1994).

The interpolation method is recognized as a key factor for interpolation accuracy. Comparison and accuracy of interpolation methods has been investigated in several studies. The reliability of the interpolation gives a better assessment of the spatial distribution of the variables (Robinson and Metternicht, 2006; Shi et al., 2009; Wu et al., 2011). Interpolation methods can be divided in three categories: (1) statistics – with regression tree and multiple regression; (2) geostatistics that include ordinary kriging and universal kriging, and (3) hybrid with co-kriging and regression kriging. Present study tested following interpolation methods that are part of the ArcGIS software: inverse distance to a weight (IDW) with the power of 1–5, local polynomial, radial basis functions (RBF)—spline with tension, completely regularized spline, multiquadratic, inverse multiquadratic and thin plate spline (TPS)—and two geostatistical methods, ordinary kriging (OK) and simple kriging (SK). They are commonly used in the literature (Erdogan, 2009; Pereira et al., 2013a,b; Xie et al., 2011). In each interpolation a total of fifteen neighbors were included and a smoothed value of 0.5 was applied.

Table 1
Particle size distribution on Stagnic Luvisols (from Kisić et al., 2002)

Soil depth (cm)	Soil horizon	Texture (g kg^{-1})				Texture class
		Coarse sand (2–0.2 μm)	Fine sand (0.2–0.02 μm)	Silt (0.02–0.002 μm)	Clay (<0.002 μm)	
0–24	Ap + Eg	18	586	242	154	Sandy loam
24–35	Eg + Btg	21	571	260	148	Sandy loam
35–95	Btg	5	545	254	196	Sandy loam

Accuracy assessment was carried out with cross-validation method previously used in papers Erxleben et al. (2002), Kerry et al. (2012), Robinson and Metternicht (2006). This method compares the observed values with the predicted ones. Cross-validation was obtained by taking each observation in turn out of the sample pool and estimating from the remaining ones. The errors produced, between observed and predicted values, are used to evaluate the performance of each method. With these data, the mean error (ME) and root mean square error (RMSE) were calculated. The ME is calculated with the following formula:

$$\mathrm{ME} = \frac{1}{N}\sum_{i=1}^{n}\left\{Z(Xi) - \hat{Z}(Xi)\right\}$$

Root mean square error is calculated following this formula:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{n}\left\{Z(Xi) - \hat{Z}(Xi)\right\}^2}$$

where $Z(Xi)$ is the observed value, $\hat{Z}(Xi)$ is the predicted value and N is the number of samples.

The best method was the one which had the lowest RMSE. Differences and correlations between observed and estimated values were identified with the paired t-test and Pearson correlation and considered significant at $p < 0.05$. Statistical analyses were carried out with SAS software (Version 9.3) and the semi-variogram and interpolation methods analysis with ArcGis 10.0 (ESRI) for Windows.

3. Results and discussion

3.1. Q–Q plots, descriptive parameters and normality test

In order to observe normality and analyze original data distribution normal Q–Q plots were created (Fig. 2). All investigated factors showed non-normal distribution. One abnormally high value was observed in AK (1090 mg kg^{-1}) at a site located close to the field entrance near channels. Apart from this extreme in AK population, only few other samples deviated at both ends. Soil AP and SOM lined up in a near straight line, but indicated some degree of left skewed distribution. A certain pattern is visible in soil pH, but the skewness is more strongly expressed. AK and pH recorded some changes in slope which are slightly more numerous than with the other factors.

Percentiles and classical descriptive parameters were calculated and are shown in Table 2. The current average contents of AP, AK, pH and SOM at 0–30 cm depth in the Pseudogley soil area were 189.1 mg kg^{-1}, 216.7 (214.0) mg kg^{-1}, 4.54 and 1.72%, respectively. Among the presented statistical data, CV value is the most discriminating factor for describing variability. CV value which is lower than 10% indicates low variability while CV value above 90% shows extensive variability (Zhang et al., 2007). All investigated parameters recorded moderate variability, while the highest variability was observed in AK with CV value of 32.64%. These results contain an outlier, but after removing that extreme data

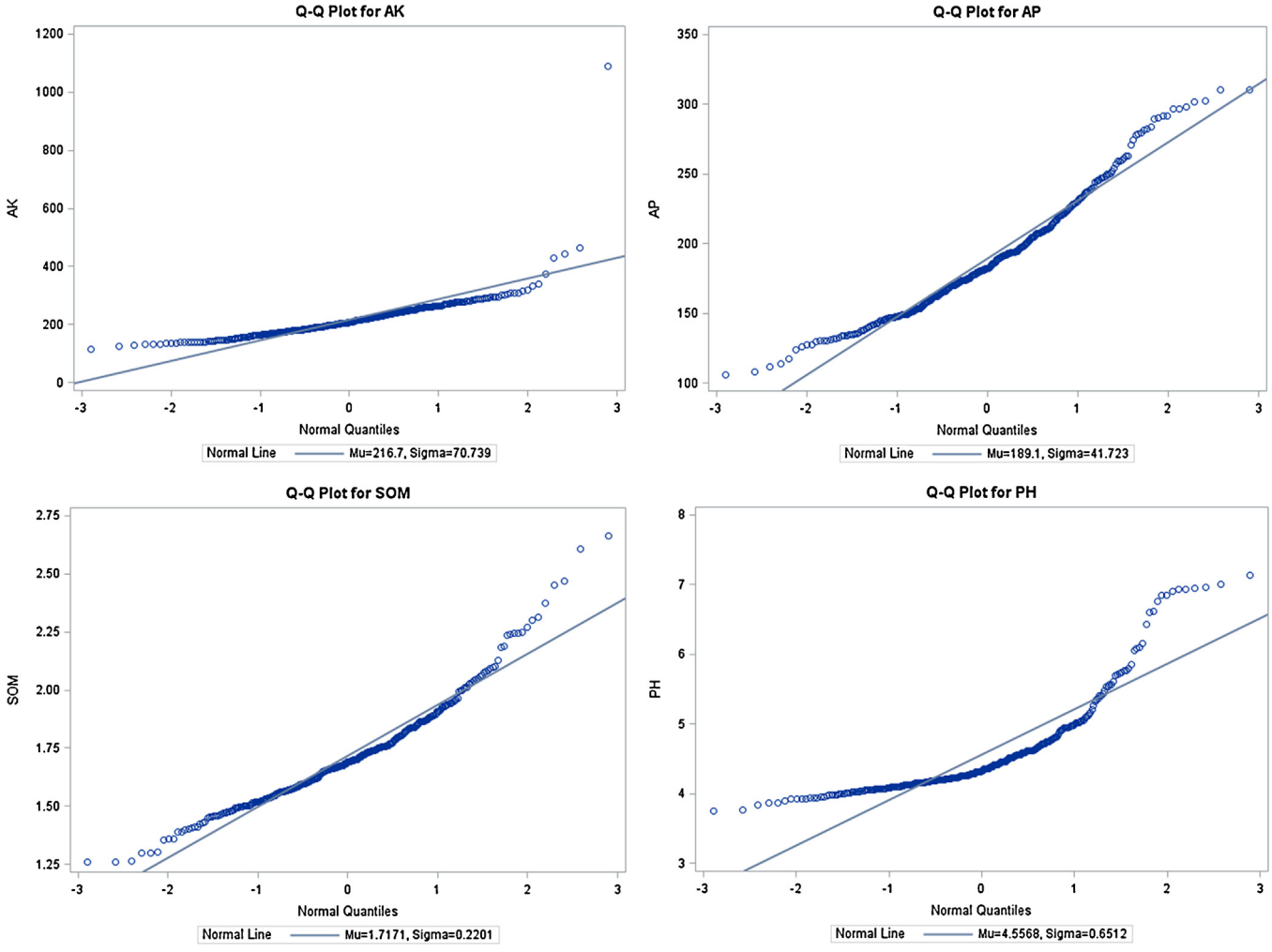


Fig. 2. Normal Q–Q plots for original data ($n = 330$). From the upper left to lower right: AK, AP, SOM and pH.

Table 2
Univariate statistics for AP (mg kg^{-1}), AK (mg kg^{-1}), pH and SOM (%).

	Min	5%	25%	Median	75%	95%	Max	Mean	CV(%)
AP	105.8	132.3	158.2	182.3	210.7	278.1	310.3	189.1	22.06
AK	115.9	139.9	176.6	206.6	249.0	294.2	1090.0 (462.7)	216.7 (214.0)	32.64 (24.21)
pH	3.75	3.98	4.15	4.33	4.71	6.05	7.14	4.56	14.29
SOM	1.26	1.43	1.57	1.69	1.83	2.10	2.66	1.72	12.82

Mean and CV values in brackets were calculated without AK outlier.

(1090 mg kg^{-1}) from the statistical analysis, CV was 24.21%, which is still higher than the CV value observed in other investigated properties.

After removing the outlier, AK skewness changed from 6.08 to 0.98 and kurtosis reduced from 70.42 to 2.49. This great change in the shape of the parameters is the reason for removing the outlier in order to obtain the characteristics of the majority of data. According to Barnett and Lewis (1994) outliers can cause distortion that violates geostatistical theory. Their influence on statistical analysis should be limited and will be removed for variogram analysis in this study. For kriging analysis outlier will be replaced with the maximum value, because an outlier can make variogram erratic (Armstrong and Boufassa, 1988). Both investigated soil nutrients (AK and AP) showed higher spatial heterogeneity in contrast to pH and SOM, but these values are far from the heterogeneity presented in papers by Fu et al. (2010), Kravchhenko (2003), Liu et al. (2009) and Schloeder et al. (2001) with CV value above 35%. Other investigated soil properties have also recorded variations. AP and AK heterogeneity could be attributed to uneven fertilizer usage or soil erosion. The CV value of SOM was only 12.82%. SOM is usually considered a relatively stable variable, with minimum variation. The SOM content in the soil usually varies and is associated with the severity of pedogenesis process. pH variations were relatively small with CV value of 14.29%. Other researchers (Castrignanò et al., 2000; Chung et al., 1995; Fu et al., 2010; McBratney and Webster, 1983; Parfitt et al., 2009) also report relatively small variations of pH of the surface layer with CV value from 2.22% (Castrignanò et al., 2000) to 8.1% (Fu et al., 2010). This relatively stabile and persistent distribution of SOM and pH is preferred for the application of lime materials and organic matter management.

The original data did not pass the K–S test for normality ($p = 0.05$). The data were highly skewed (Table 3), and this can be observed in the shape of Q–Q plots (Fig. 2). After the log transformations, data confirm the Gaussian distribution for AP and AK, while pH and SOM did not follow normal distribution. After the Box–Cox transformation data were closer to the Gaussian distribution, but still they did not follow the normality in pH content, while SOM content followed a normal distribution. However, values of pH data after Box–Cox transformation in contrast to previous transformations, were closer to the normal distribution, and the skewness and kurtosis were closer to 0. This situation is not unheard of and is evidenced in McGrath and Zhang (2003) study. They observed that after log data transformation, pH values did not pass the K–S test, but authors modeled the variable with log transformed data. In this study, pH dataset showed smaller skewness and kurtosis after Box–Cox transformation and these data were used for modeling. If data cannot be modeled with a normal distribution, they should at least be as close to the normal distribution as possible (Zhang, 2006). Pereira et al. (2013a,b) follow a similar procedure in studies with ash thickness modeling. Therefore, the log-transformed data sets of AP and AK, and the Box–Cox transformed data sets of SOM and pH were used for the following multivariate analyses.

3.2. Spatial characteristics of investigated properties

The parameters for variogram models are listed in Table 4 and the fitted variogram plots for AK, AP, SOM and pH are shown in Fig. 3. All variogram models recorded directional differences which indicate differently increased variogram values at different directions. The longest range spatial dependence for AP, AK, SOM and pH was found in directions 40, 25, 85 and 70, respectively. Among other models, the exponential model gave the best fit for the experimental variograms calculated for AP, AK and pH (Fig. 3a, b and d). Nugget effect was not recorded for soil AP while the recorded nugget for AK and pH was 0.00427 and 0.00001, respectively. Compared to sill, small nugget values indicate that sampling errors are negligible. Usually the nugget effect occurs as a consequence of limited sample numbers, small-scale variance and the existence of outliers (Gringarten and Deutsch, 2001; McGrath and Zhang, 2003). In present study, the number of samples was representative of the studied plot, and the outlier in AK was replaced with the maximum value. Thus, the nugget effect was attributed to the small-scale variance observed in some areas of the plot in the current sampling density.

AP and pH variable with nugget/sill ratio of 0.00 and 19.54, respectively, showed strong spatial dependence, while AK with a 36.33 nugget/sill ratio showed moderate spatial dependence. The spherical model fitted to the experimental variogram calculated for SOM data (Fig. 3c) recorded a nugget effect of 0.00223 and a sill of 0.00320 along a range of 864.0 m. The nugget/sill ratio (69.56) showed that SOM had a moderate spatial dependence. Ranges of all investigated parameters were much wider than the sampling interval of 50 m. According to the geostatistical theory, this sampling design is sufficient for the investigation of spatial dependence and distribution of investigated parameters in the investigated soil.

Kerry and Oliver (2004) indicated that a rough guide for choosing future sampling interval should be less than half the

Table 3
Shape parameters and K–S test for original and transformed data.

	Original data			Log-transformed data			Box–Cox transformation			
	Skewness	Kurtosis	K–S p	Skewness	Kurtosis	K–S p	Lambda	Skewness	Kurtosis	K–S p
AP	0.71	0.21	<0.01	0.18	−0.29	>0.15				
AK	0.98	2.49	<0.01	0.14	−0.01	>0.15				
pH	2.03	4.29	<0.01	1.64	2.66	<0.01	−2	0.98	0.55	<0.01
SOM	1.05	2.09	<0.01	0.55	0.90	<0.01	−1.5	−0.13	0.51	>0.15

Parameters for AK were calculated without the outlier.

Table 4
Best-fitted anisotropic variogram models of AP, AK, SOM and pH and corresponding parameters.

	Model	Nugget C_0	Partial sill C	Sill C_0+C	Nugget/Sill	Range (meters)
AP	Exponential	0.00000	0.00718	0.00718	0.00	201.3
AK	Exponential	0.00427	0.00749	0.01177	36.33	1153.6
SOM	Spherical	0.00223	0.00097	0.00320	69.56	864.0
pH	Exponential	0.00001	0.00003	0.00004	19.45	1954.4

variogram range. According to the results of this study future sampling grid should be at least 100 m for AP and 400 m for monitoring AK and SOM. Suitable sampling interval of 100 m for AP is also suggested in papers by Mallarino et al. (2006) and Fu et al. (2010) in 30 m grid intervals and 150 m interval is suggested for AK (Fu et al., 2010). Soil pH shows incredible high range (1954.4 m) and significant spatial dependence (nugget/sill 19.45). Soil of the investigated area has naturally low pH regardless of the environmental and topographical factors. The results of this study showed that 90% of samples are very acid (pH < 4.5) and acid (pH from 4.5 to 5.5) and this is the reason for high spatial homogeneity of soil pH that resulted in high range in variogram. This pH value is a great warning and indicates the limitations of these soils for agricultural production. In future production period significant amount of liming materials will be needed to improve soil pH.

Current soil sampling strategy in Croatia is unclear, insufficient for the continuous monitoring of soil and is performed by only a small number of corporations. It is mostly done as one representative sample made by 20 soil cores in a zigzag pattern from a 5 ha field area. According to the results of this study this sampling method may not be adequate for proper agricultural soil monitoring and does not provide enough accurate information for fertilizer and lime application. Taking into account the constant problems with the cost of precise sampling, sampling scheme on 100 m to a maximum of 150 m, depending of the soil type, past land use and topographic factors, can be recommended. Although corresponding parameters of variogram models suggests a greater distance for samples, these must be taken into consideration with caution because some important soil attributes may be missing under natural soil variability when the grid size is too large.

3.3. Interpolation methods assessment and spatial distribution

The results of the tested interpolation methods for all parameters are shown in Table 5. The test of the different interpolation methods provides an accurate insight in the spatial distribution of AP, AK, SOM and pH on sandy loam soil. CRS method was the most accurate for interpolating the AP (RMSE, 0.0540) and the least precise method was IDW1 (RMSE, 0.0812). The most precise method for AK was IMTQ (RMSE 0.0570) and the least accurate method was IDW1 (RMSE 0.0879). LP1 proved to be the most accurate method for SOM (RMSE 0.0473) and the least precise

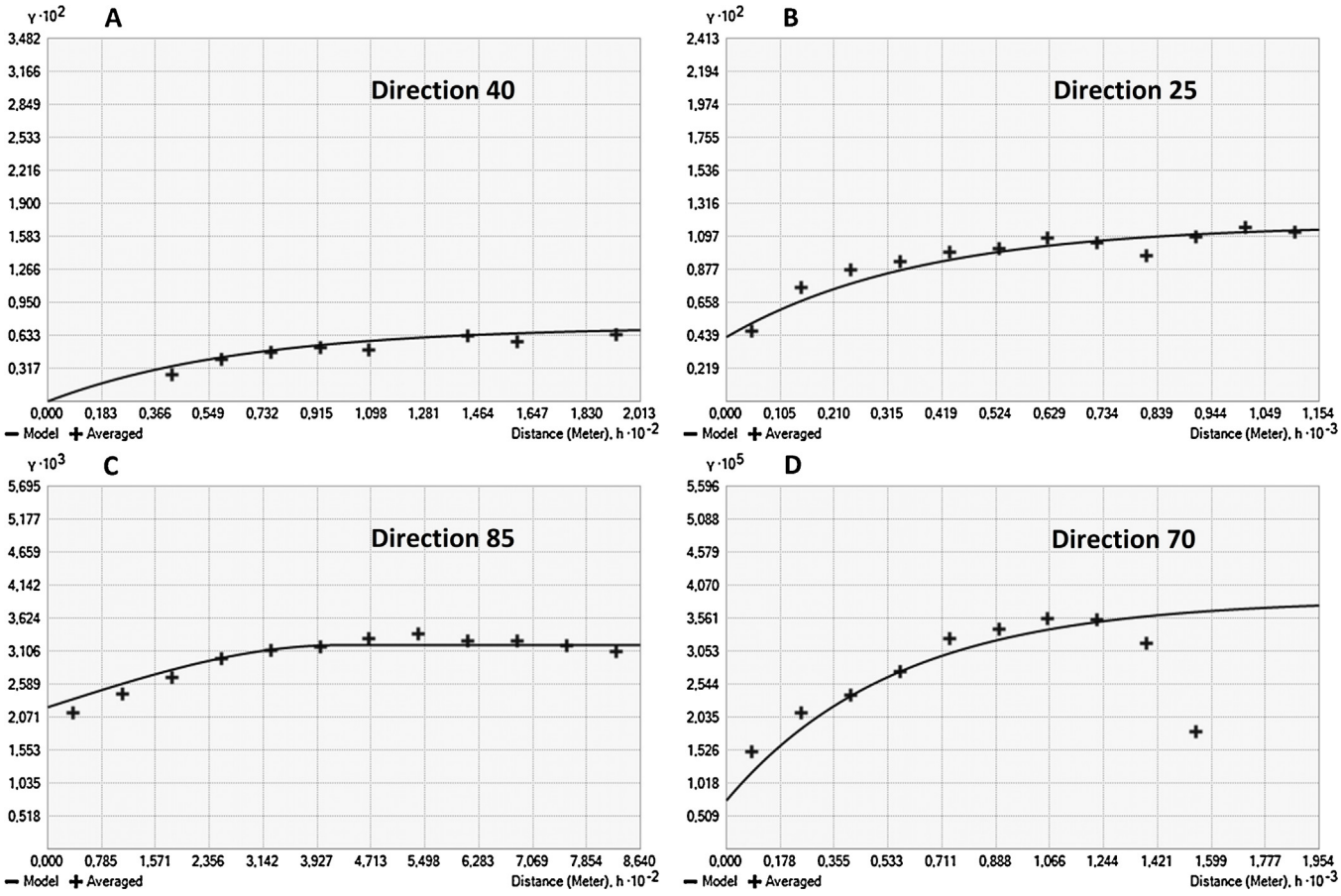


Fig. 3. Directional semi-variogram calculated for AP (a) and AK (b) with log-transformed data; SOM (c) and pH (d) with Box–Cox transformed data. Bold line represents the best fit model.

Table 5
Summary statistics of the accuracy of interpolation methods.

AP/model	Min	Max	ME	RMSE	*r*	SOM/model	Min	Max	ME	RMSE	*r*
IDW1	−0.207	0.224	−0.00464	0.0812	0.55^{ns}	IDW1	−0.128	0.176	−0.00045	0.0505	0.37^{ns}
IDW2	−0.187	0.197	−0.00406	0.0725	0.67^{ns}	IDW2	−0.130	0.183	−0.00031	0.0495	0.41^{ns}
IDW3	−0.186	0.190	−0.00384	0.0655	0.73^{ns}	IDW3	−0.134	0.192	−0.00032	0.0500	0.40^{ns}
IDW4	−0.181	0.187	−0.00354	0.0621	0.75^{ns}	IDW4	−0.147	0.202	−0.00032	0.0514	0.37^{ns}
IDW5	−0.177	0.188	−0.00324	0.0610	0.76^{ns}	IDW5	−0.157	0.209	−0.00032	0.0528	0.35^{ns}
LP1	−0.175	0.224	0.00607	0.0588	0.78^{ns}	LP1	**−0.139**	**0.174**	**0.00089**	**0.0473**	**0.49**^{ns}
LP2	−0.204	0.201	−0.00187	0.0574	0.80^{ns}	LP2	−0.177	0.227	0.00042	0.0562	0.34^{ns}
SPT	−0.173	0.194	−0.00067	0.0547	0.75^{ns}	SPT	−0.141	0.196	−0.00019	0.0501	0.41^{ns}
CRS	**−0.172**	**0.193**	**−0.00152**	**0.0540**	**0.75**^{ns}	CRS	−0.149	0.203	−0.00014	0.0511	0.38^{ns}
MTQ	−0.173	0.200	−0.00092	0.0543	0.80^{ns}	MTQ	−0.181	0.227	0.00012	0.0564	0.30^{ns}
IMTQ	−0.230	0.212	−0.00158	0.0541	0.63^{ns}	IMTQ	−0.130	0.179	−0.00005	0.0498	0.39^{ns}
TPS	−0.196	0.199	−0.00067	0.0547	0.81^{ns}	TPS	−0.214	0.252	0.00027	0.0653	0.22^{ns}
OK	−0.180	0.199	−0.00176	0.0564	0.80^{ns}	OK	−0.132	0.183	−0.00018	0.0486	0.44^{ns}
SK	−0.191	0.181	−0.00247	0.0569	0.79^{ns}	SK	−0.135	0.175	−0.00029	0.0482	0.46^{ns}

AK/model	Min	Max	ME	RMSE	*r*	pH/model	Min	Max	ME	RMSE	*r*
IDW1	−0.264	0.252	−0.00037	0.0879	0.58^{ns}	IDW1	−0.015	0.011	−0.00033	0.00435	0.65^{ns}
IDW2	−0.214	0.208	−0.00099	0.0762	0.71^{ns}	IDW2	−0.013	0.012	−0.00020	0.00372	0.75^{ns}
IDW3	−0.201	0.176	−0.00164	0.0684	0.76^{ns}	IDW3	−0.013	0.013	−0.00011	0.00338	0.78^{ns}
IDW4	−0.209	0.184	−0.00179	0.0649	0.78^{ns}	IDW4	−0.012	0.014	−0.00007	0.00330	0.79^{ns}
IDW5	−0.223	0.191	−0.00170	0.0635	0.79^{ns}	IDW5	−0.012	0.014	−0.00006	0.00332	0.78^{ns}
LP1	−0.200	0.136	0.00211	0.0602	0.81^{ns}	LP1	−0.011	0.012	0.00020	0.00313	0.81^{ns}
LP2	−0.200	0.136	−0.00033	0.0606	0.81^{ns}	LP2	−0.012	0.013	−0.00013	0.00320	0.81^{ns}
SPT	−0.219	0.150	−0.00075	0.0585	0.83^{ns}	SPT	−0.010	0.014	0.00001	0.00297	0.83^{ns}
CRS	−0.227	0.166	−0.00089	0.0573	0.83^{ns}	CRS	**−0.011**	**0.015**	**−0.00002**	**0.00285**	**0.85**^{ns}
MTQ	−0.229	0.162	−0.00078	0.0575	0.83^{ns}	MTQ	−0.012	0.015	0.00002	0.00286	0.85^{ns}
IMTQ	**−0.222**	**0.159**	**−0.00044**	**0.0570**	**0.84**^{ns}	IMTQ	−0.011	0.015	−0.00002	0.00301	0.83^{ns}
TPS	−0.219	0.149	−0.00078	0.0584	0.83^{ns}	TPS	−0.010	0.014	0.00001	0.00297	0.83^{ns}
OK	−0.210	0.156	−0.00040	0.0649	0.78^{ns}	OK	−0.012	0.014	−0.00002	0.00322	0.80^{ns}
SK	−0.207	0.164	−0.00077	0.0616	0.81^{ns}	SK	−0.012	0.014	−0.00007	0.00317	0.81^{ns}

Numbers in bold font indicate the most accurate method. Correlations between observed and estimated values is significant at $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$ and ns: not significant.

method was TPS (RMSE, 0.0653), while for soil pH the most accurate method was CRS (RMSE, 0.67) and the least precise method was IDW1 (RMSE, 0.00435). Among the factors that mostly affect mapping accuracy is the number of samples, the distance between sampling locations and the choice of interpolation method (Kravchenko, 2003). Tested methods showed that the ME was very close to 0 in all parameters which suggests that predictions are unbiased. Also, there is a very small difference between observed and predicted values which results with no significant differences between the correlation coefficient of observed and predicted values. This pattern occurs on all investigated parameters. The predictions do not deviate much from the measured values, as seen in RMSE and ME which are as small as possible. Generally, a larger number of samples will produce more accurate spatial map (Mueller et al., 2001) and the results are a consequence of sufficient number of samples in this study. Correlation coefficient between observed and predicted values (Table 5) was not significant in all tested methods on all parameters. Their calculated errors are random and not justified which goes in favor of the precision and accuracy of tested methods. All correlation coefficients showed positive correlation and were in range of 0.49 (SOM)–0.85 (pH), while the highest correlations were observed for pH and AK that recorded the highest spatial dependence and ranges of 1954.4–1153.6 m, respectively. The attention needs to be focused only on SOM maps, because their samples do not have strong spatial dependence as other investigated parameters (Table 4).

In order to facilitate fertilizer, organic matter and lime application, the nutrient, SOM and pH index system (Fig. 4) was adopted to mark the field map based on the present distribution of concentrations. Higher AP and AK concentrations were recorded at the similar areas on south-southwest part of the field and also near a channel that divides northeast part of the field. The area near this channel also recorded higher pH concentrations due to the changes in natural soil horizons. During channels excavations the soil from deeper layers scatters the surrounding area and this process changes soil characteristics, resulting in a higher concentration of accessible nutrients. In the same way the increased concentration of accessible nutrients to all end-field areas can be explained. Central parts of the field recorded lower pH and it is visible that concentrations of AK and AP follow the changes of soil pH. High acidity accompanied by low cation-exchange capacity causes problems when these soils are utilized in agriculture. Nutrient deficiency and high levels of exchangeable aluminum are the main problems (Macías et al., 2008). In general, an increase in soil acidity is associated with a decrease in the availability of nutrient ions: particularly calcium, magnesium, phosphorus, and molybdenum (Ritchie, 1989; Robson and Abbott, 1989). Availability of phosphorus in soil is the greatest at pH range of 5.5 to 7.0 (Mesic et al., 2013). Positive influence of liming and increase of soil pH in relation to AP in soil was noted by other authors (Rastija et al., 2008, 2010). Rahman et al. (2002) and Zhang et al. (2004) elaborated the influence of liming on phosphorus availability in acid soils, stressing that a moderate pH increase leads to greater phosphorus availability, while too high doses can lead to its decreasing. Additional sources of acidity that release hydrogen (H^+) to the soil solution include oxidation-reduction or redox reactions, addition of soluble salts trough fertilizers, and base cation (Ca, Mg, K) removal during plant uptake (acidification of the rhizosphere) (Thomas and Hargrove, 1984). According to the information gained from a personal communication with the owner of the company, significant amounts of nitrogen fertilizer were applied on the investigated areas. The use of larger amounts of nitrogen fertilizer and natural acidification of soils due to constant de-basification results in reduced fertility of the whole field because of higher acidity. Paying more attention to liming than the fertilization in future management of investigated field is strongly recommended, because AP and especially AK in soil was sufficient and no more

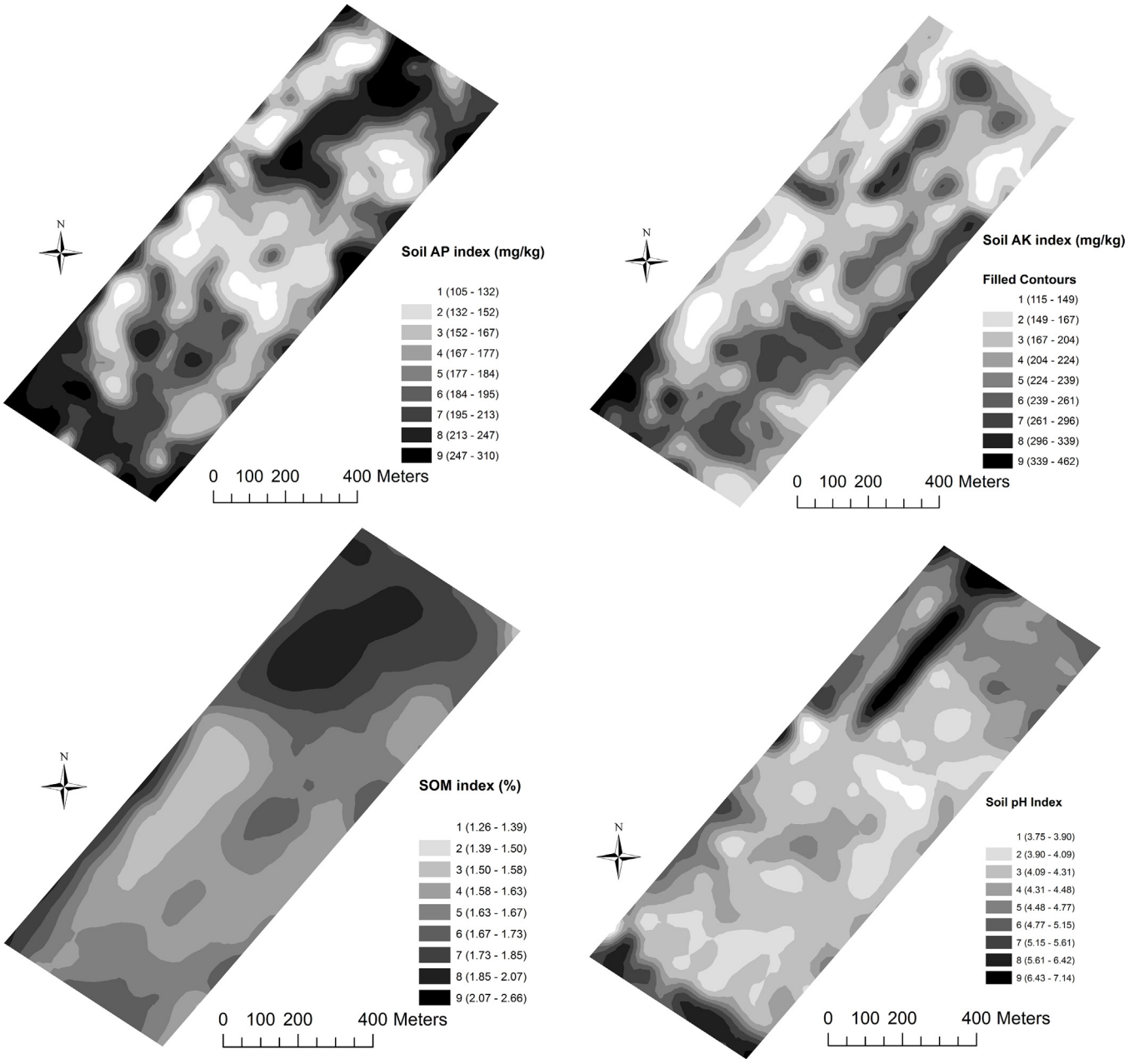


Fig. 4. Spatial distribution maps according to the most accurate technique. Left-to-right: first row: AP (CRS) and AK (IMTQ) with log-transformed data. Second row: SOM (LP1) and pH (CRS) with Cox–Box transformed data.

phosphorus and potassium fertilizer is needed. Concentrations of SOM are the most favorable on the northern part of the field and near the field edges, more or less following the spatial disposition of pH. SOM concentration strongly depends on plant residues management, crop rotation, climate, nitrogen availability and microbiological activity in soil. Tillage operations and climate are the same at the whole field and thus a conclusion can be drawn that the pH, indirectly through soil conditions, directly affects the microbial activity and regeneration of humic particles.

4. Conclusions

Investigated parameters do not follow normal distribution. The application of log–and Cox–Box transformations was effective in shifting the data sets to normality with acceptable skewness values. For AP, AK and pH best variogram fitted model was exponential. The spatial structure of SOM was fitted by spherical model. Practical application of research results comes from the fact that the inclusion of established models directional semivariograms in interpolation analysis can improve the reliability of local assessments of the analyzed soil properties at unsampled locations, thus reducing the cost of the production cycle. All investigated parameters (AP, AK, SOM and pH) recorded higher range than sampling interval with 201.3, 1153.6, 864.0 and 1954.4 m, respectively. In order to reduce production costs a sampling interval of 100 to 150 m is recommend, depending on the parameter. However, since all of the parameters are measured from the same soil samples, the recommended interval is 100 m. All tested interpolation methods have high prediction accuracy of the mean content. Models of spatial distribution of AP, AK and pH datasets evaluated by RBF (CRS, IMTQ and CRS) models provide a more realistic picture of the structures of analyzed spatial variables

in contrast to kriging and IDW models. For SOM datasets the most favorable model was LP1. This indicates that prior to the implementation of the development of distribution maps, parameters of semivariograms need to be computed. These results lead to a conclusion that non interpolator can produce results for continuous monitoring of certain soil property. All investigated models provide small RMSE and ME but it is necessary to achieve the smallest error in prediction to avoid economic cost. The use of interpolation methods and geostatistics proved a possibility to achieved enough accurate fertility maps to provide correct information for site specific fertilization and liming application. This will in turn result in reduced use of fertilizers, favorable soil reaction, increased soil microbial activity and will reduce potential environmental problems.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.still.2014.07.020.

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